You-Zhe Xie (謝侑哲)

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Transcript | Google Scholar | linkedin | github | Twitter

Education

National Yang Ming Chiao Tung University, BS in Computer Science 09.2023 –06.2027

- Ranking: **1/197 (0.51%)**, GPA: 4.29/4.30
- **Language**: English (TOEIC 810), Chinese (Native)

Experience

Research Intern, Alaya Lab 12.2025 - Now

- Research in **video generation models' understanding of causality** (YoCausal). Built the first benchmark for causal cognition in video generation models, extensible with arbitrary real video at zero cost. 1 first-author paper in submission to ICLR 2027.
- Building a benchmark for **extracting game mechanics from gameplay videos with MLLMs**.

Research Assistant, CPLab, NYCU (Advisor: Prof. Yu-Lun Liu) 06.2024 - Now

- Research in **3DGS poisoning attack** (StealthAttack). Stealthily inject illusory objects into a target view of 3DGS, achieving **SOTA** in this task. Published **1 ICCV paper**; delivered an **oral presentation** and won the **Best Poster Award** at an ICCV workshop.
- Research in **3D cinemagraph generation** (OuroWorld). Animate any static 3D scene into a seamlessly looping 3D cinemagraph. 1 first-author paper in submission to ICLR 2027.
- Research in **MLLM-guided physical parameter estimation from video** (TwinReason). 1 paper in submission to WACV 2027.

Research Intern, AIIULab, Academia Sinica (Advisor: Dr. Jun-Cheng Chen) 07.2025 - Now

- Research in **4D Gaussian ray tracing** for **physics-based, fine-grained camera effect data generation** (4DGRT). Achieve **highest data generation speed** and comparable quality to NeRF-based method.
- Published **1 NeurIPS workshop paper** and delivered an **oral presentation**.

Consultant, WorldQuant 02.2025 - Now

- Design strategy for **quantitative trading** to allocate investment funds. The best is to achieve **average 13.2% returns per year**.

Research Assistant, Department of Management Science, NYCU 02.2025 - 06.2026

- Research in the Lotka-Volterra model based on biological competition for sales forecasting. Analyze competitive relationships between different products.

Mentor, Google Developer Student Club, NYCU 09.2024 - 06.2025

- Taught courses on **AI and computer vision** and advised projects about **font style transfer** & **weakly supervised semantic segmentation**.

Publications

YoCausal: How Far is Video Generation from World Model? A Causality Perspective 03.2026
You-Zhe Xie, Yu-Hsuan Li, Jie-Ying Lee, Kaipeng Zhang, Yu-Lun Liu, Zhixiang Wang
ICLR 2027 (Under Submission)

OuroWorld: Bringing Any 3D World Alive as Diverse, Endlessly Looping 3D Cinemagraphs 09.2026
You-Zhe Xie, Yu-Hsuan Li, Ting-Wei Chou, Zhixiang Wang, Yu-Lun Liu
ICLR 2027 (Under Submission)

StealthAttack: Robust 3D Gaussian Splatting Poisoning via Density-Guided Illusions 03.2025
Bo-Hsu Ke, ***You-Zhe Xie (2nd)***, Yu-Lun Liu, Wei-Chen Chiu
ICCV 2025 (Poster)
ICCV 2025 Responsible Imaging Workshop (★ Oral, ✹ Best Poster)

Every Camera Effect, Every Time, All at Once: 4D Gaussian Ray Tracing for Physics-based Camera Effect Data Generation 08.2025
Yi-Ruei Liu, ***You-Zhe Xie (co-1st)***, Yu-Hsiang Hsu, I-Sheng Fang, Yu-Lun Liu, Jun-Cheng Chen
WACV 2027 NeurIPS 2025 SpaVLE Workshop (★ Oral)

TGPED: Task-Generalized Procedural Error Detection and Explanation using Large Language Models with Procedural Grounding 09.2026
Siangyu Yang, S-Wei Chen, ***You-Zhe Xie (3rd)***, Ho-En Tsai, Yu-Chi Lai, Kai-En Peng, YouChen Lai, Yu-Lun Liu, Chun-Yi Lee, Shih-Po Lee, Shao-Yuan Lo
ICLR 2027 (Under Submission)

TwinReason: Cross-Engine MLLM-Guided Inverse Physics from Monocular Video 03.2026
Chih-Hai Su, Fu-En Wang, Nan Qiao, Yuyin Sun, ***You-Zhe Xie (5th)***, Yang-Sen Lin, Jie-Ying Lee, Yi-Ruei Liu, Roberto Martín-Martín, Yu-Lun Liu, Tsung-Wei Ke, Min Sun, Cheng-Hao Kuo

Projects

ShellBench: A Parametric Shell Game Benchmark for Memory-Aware Manipulation 06.2026

- ShellBench is a parametric **Isaac Sim** shell-game benchmark that measures the **long-term memory** capacity of visuomotor

manipulation policies by forcing them to track a ball hidden under shuffled identical cups. Cup count, shuffle count and speed are tunable, yielding memory-capacity curves that quantify each model's **memory decay**.

Designing Fun: LLM-based Game Level Generation via Agent Gameplay Feedback 06.2026

- A two-stage SFT + **GRPO** framework that trains an LLM to generate fun MiniGrid levels by optimizing three gameplay-grounded rewards: regret, interaction, and diversity. Outperforms vanilla and SFT-only baselines on all metrics.

MATCHA: Match with Agent 04.2026

- Built at the 2026 YTP Hackathon. MATCHA is a **multi-agent** system where personal AI **Agents socialize** on users' behalf, using **RAG** to actively match them with government career resources and peer experiences.

Desktop Organizer 12.2025

- Built an intuitive desktop organizer based on **user research** to ensure ease of use. Integrated **LLM** for automatic file management, designing a simple **React** interface that makes managing desktops effortless.

Rewinded Video Revision 12.2025

- Revise the background artifact of videos with rewind effect. Solve the problem by 3 stages: foreground **segmentation**, unreasonable **region detection**, **tuning-free video revision**. This project later **inspired YoCausal**.

Real-Time DSTNet 08.2024

- Improved **video deblurring** model DSTNet to **real-time inference** (37 fps), **sped up to 400%**, refactored from offline to online.

Anime Helper 12.2024

- Anime info. website with **React.js**, **Express.js**, **PostgreSQL**. I was responsible for the **backend** engineering.

Paper Searching Bot with LLM 06.2024

- Built paper searching discord bot with **RAG** on LLM. Optimized result by prompt engineering. Collected data via crawler. Achieved **96.6% content relativity**.

Deep Learning Projects 08.2024

- 6 sub-projects with technique including: CNN, UNet, ResNet, VAE, Transformer, Diffusion etc.

Berkeley Computer Vision Projects 08.2024

- 5 sub-projects include image alignment, hybrid, warping, stitching, auto stitching.

Honors

First Place (特優), NYCU Undergraduate Project Competition : Awarded for YoCausal.

Academic Excellence Award (書卷獎) * 5 : Top 5% in the class.

Core Fundamental Course Award (核心基礎課程獎) * 4 : Top 5% in specific courses.

- Algorithm, Discrete Mathematics, Digital Circuit Design, Computer Organization